

Aric Nguyen

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SUMMARY

Data Engineer / Solution Architect with 4 years building and operating production data platforms across Microsoft Fabric, Azure, Databricks, and open-source lakehouse stacks. Hands-on across the full data product lifecycle — architecture decisions and ADRs, batch and real-time pipelines, hybrid Lakehouse + Warehouse design, business semantic layers, automation and data acquisition, CI/CD, and governance. Comfortable owning end-to-end delivery in cross-border environments (APAC ↔ US HQ); equally fluent translating engineering trade-offs to business stakeholders.

EXPERIENCE

Ashley Furniture Industries

Binh Duong, Vietnam · Onsite (US HQ-aligned)

Data Engineer / Solution Architect

Jan 2026 — Present

Led architecture refactor of global supply-chain analytics platform on Microsoft Fabric (5,000+ enterprise tables, 30 years of history, billions of records) from a Spark-based pipeline to a warehouse-native Hybrid Medallion. Cross-border alignment with US HQ technical reviewers; mentored local team on platform patterns.

- **Designed a metadata-driven control plane** separating platform concerns (registry, generic load runner, lineage builder, DQ engine, audit, scheduler, semantic-refresh) from data layers — onboarding a new dataset becomes a registry insert, with zero per-asset code or stored-procedure changes.
- **Built a generic load framework covering 8 patterns** (overwrite, incremental, upsert, SCD2, CDC, date-key, date-range, identity) routed through metadata; replaced the prior per-table SP pattern entirely while preserving idempotency, watermark integrity, and replay safety.
- **Cut end-to-end mart cycle from ~60 min to ~20 min** through DAG-based parallel execution with automated dependency resolution, cron-driven scheduling with smart-skip for stale-ineligible assets, and multi-mart routing that orchestrates N marts in parallel by registry partition.
- **Engineered the Direct Lake semantic boundary** — physical-Gold tables (per official Microsoft guidance) prevent silent fallback to DirectQuery; semantic-model refresh orchestrated post-load via Fabric REST API, surfacing latest data within minutes of ETL completion.
- **Established Data Quality & schema contracts as preflight gates** — drift detection and severity-tiered DQ checks halt the pipeline on critical failures (CRITICAL halts, WARNING logs); coverage scales automatically with each new asset, and source-target reconciliation prevents silent source drift on shortcut reads.
- **Built internal architecture tooling** (stability scanner, registry health checks, readiness baseline exporter, Streamlit-based Lineage Explorer on top of governance metadata) to make platform assessment repeatable rather than ad-hoc.
- **Operationalized production CI/CD** — Azure DevOps with .sqlproj (DacFx) build validation, schema-diff publish via SqlPackage, PR review gates, SqlCommandVariable-driven multi-environment promotion (DEV → TEST → PROD), and a side-by-side non-destructive build pattern keeping v9 fully alive until v10 passes parity and approval gates.
- **Reached ~91% compatibility with US HQ enterprise warehouse patterns** validated by US HQ technical reviewers; delivered DAG orchestration, auto-built lineage, multi-type DQ gates, semantic auto-refresh, and multi-mart readiness as capabilities *beyond* the US HQ baseline.

Ecentric

Ho Chi Minh City, Vietnam

Data Engineer / Analytics Engineer

Mar 2023 — Jan 2026

End-to-end owner of the company's data and analytics platform on Microsoft Fabric and Azure, including phased migration from an Azure stack to a unified hybrid Lakehouse + Warehouse architecture. Delivered production pipelines, real-time streaming, automation and scraping, business semantic layer, and AI-powered BI serving 4 departments and 15+ Power BI reports — reducing manual reporting effort by 70%+.

- **Architected a hybrid Lakehouse + Warehouse platform on Microsoft Fabric** — Lakehouse for raw/structured ingestion and PySpark transforms; Warehouse for governed serving and DAX-friendly star schemas; OneLake shortcuts between workspaces enable zero-copy data sharing across domains.
- **Led phased zero-downtime migration from Azure (ADF + Azure SQL DB) to Fabric** — schema mapping, parallel-run validation, cutover planning, plus production Mirroring from Azure SQL DB → Fabric for near-real-time replication; documented decision rationale (unified compute, lower TCO, Direct Lake removing import-refresh bottleneck) in ADRs.
- **Built end-to-end real-time streaming pipelines** — owned architecture, ETL, and ops: event ingestion, transformation, and sink-out to analytical layers; co-existed with batch medallion for unified batch + streaming serving (lambda-style).
- **Engineered a metadata-driven medallion engine in PySpark** across Bronze → Silver → Gold with watermark-based incremental loads, merge/upsert for SCDs, schema-on-read at Bronze with typed enforcement at Silver, and 3-level blast-radius isolation (table → domain → platform) for safe parallel execution.
- **Optimized Spark and Delta Lake at scale** — partition tuning, Z-order & V-order optimization, scheduled VACUUM / OPTIMIZE jobs, predicate pushdown, broadcast joins for skew, and Fabric capacity / CU monitoring with cost-aware workload scheduling.
- **Built a comprehensive automation & data acquisition layer** — REST API integrations across diverse sources (CRM, ERP, marketing, finance, e-commerce sandboxes) with auth handling (OAuth2, API keys, signed requests), retry logic, rate-limit backoff, and pagination patterns; Power Automate (Desktop + Cloud) for office-process automation; self-hosted n8n for cross-system orchestration; Playwright/Selenium-based

web scraping with anti-bot resilience; multi-marketplace e-commerce integrations (Shopee / Lazada / Tiki) for orders, products, and inventory ingestion.

- **Programmatic platform automation via REST APIs** — used Fabric REST API, Power BI REST API, and Azure Management API to provision workspaces, deploy items, manage permissions, trigger semantic refreshes, and rotate credentials as code; treated platform configuration as repeatable artifacts rather than click-ops, integrated with CI/CD for environment promotion.
- **Designed an enterprise-grade business semantic layer on Power BI** — Tabular Editor-built SSAS-style models with hierarchies, KPIs, and calculation groups; complex DAX measures (time intelligence, advanced CALCULATE filter context, virtual relationships); RLS / OLS / CLS for row-, object-, and column-level security across Sales, Finance, HR, and Service.
- **Architected an internal layered analytics platform** with three-layer separation (transformation SQL in Git-tracked dictionaries, execution config in metadata, reusable PySpark modules); domain-parallel execution with multi-checkpoint DQ framework, automated halt on critical failures, and self-populating data dictionary with full lineage traceability.
- **Delivered Direct Lake BI & AI capabilities** — 15+ Power BI reports across Sales / Finance / HR / Service with conformed dimensions and measures (GMV, conversion, marketing attribution, cohort retention, basket analysis); custom Deneb (Vega-Lite) and HTML visuals; co-developed an internal analytics web app (React + Streamlit); shipped MCP-based conversational AI for DAX generation and report scaffolding; integrated Azure OpenAI into the Fabric ecosystem.

TheSocietyPass (SOPA Vietnam)

Remote (Contract)

Data Engineer

Mar 2023 — Jan 2024

Co-owned architecture decisions and infrastructure operations on a streaming-capable data platform for multi-marketplace e-commerce analytics, alongside a senior data engineering team.

- **Implemented multi-pattern ingestion** — batch APIs (Airbyte), CDC (Debezium), WebSocket / webhook receivers — converged into Kafka / Redpanda with a schema registry for evolution and backward compatibility.
- **Built Flink SQL stream processing** for real-time transforms (windowed aggregations, sessionization, anomaly detection); co-managed job lifecycle, checkpointing, and state backend configuration.
- **Built batch medallion on Iceberg lakehouse** with dbt + Trino as query engine; dual-write to ClickHouse (hot, sub-second OLAP) and Iceberg / MinIO (cold, long-term) for tiered access patterns.
- **Co-managed infrastructure operations** — Dockerized deployments, Kafka cluster ops, Trino / Iceberg catalog config, Terraform-provisioned infrastructure, monitoring and alerting; built a feature store with batch + streaming features for DS self-service.

ADP

Remote

Data Engineer (promoted from Data Analyst Intern in 3 months)

Jan 2022 — Nov 2022

- **Built dbt transformations on PostgreSQL** with incremental models, snapshot strategies, and data-quality tests across HR / payroll domains (headcount, turnover, payroll variance, compliance).
- **Configured Airflow DAGs** for daily / weekly batch processing with SLA-breach and pipeline-failure alerting across multiple sources.
- **Deployed OpenMetadata for data governance** — catalog, lineage, and DQ scoring; established data-dictionary standards, naming conventions, and source-of-truth documentation across teams.
- **Developed Superset dashboards and Power BI reports** for HR analytics; optimized PostgreSQL schemas for analytical workloads with indexing strategy and materialized views.

SKILLS

Languages: Python; SQL (T-SQL, Spark SQL, PostgreSQL); DAX; M (Power Query); JavaScript

Microsoft Cloud & Fabric: Microsoft Fabric (Lakehouse, Warehouse, Pipeline, Dataflow Gen2, Spark Notebook, Semantic Model, Mirroring, OneLake Shortcuts); Azure (Data Factory, Synapse, SQL DB, Data Lake Storage Gen2, Functions, Event Hub, Key Vault, Entra ID, DevOps); Delta Lake; Direct Lake; Databricks (notebooks, Unity Catalog, Delta Live Tables, Workflows)

Data Engineering & Streaming: PySpark; dbt; medallion architecture; MERGE / Upsert; SCD2; CDC; incremental load + watermark; schema validation; dimensional modeling; lineage; Spark partition / Z-order / V-order tuning; Delta VACUUM / OPTIMIZE; Kafka / Redpanda; Flink; Debezium CDC; webhook / WebSocket ingestion; schema registry

API & Automation: REST API design and consumption; Fabric REST API; Power BI REST API; Azure Management API; OAuth2 / API key / signed-request auth; pagination, rate-limit backoff, retry patterns; Power Automate (Desktop + Cloud); n8n (self-hosted); Playwright / Selenium / BS4 web scraping; e-commerce platform APIs (Shopee / Lazada / Tiki)

DevOps & CI/CD: Git; Azure DevOps; DacFx / .sqlproj / SqlPackage; GitHub Actions; Service Principal / Entra ID; Docker; Terraform

BI & Semantic Layer: Power BI (Direct Lake, DirectQuery, Import); Tabular Editor (SSAS-style models, hierarchies, KPIs, calc groups); complex DAX (time intelligence, CALCULATE filter context); RLS / OLS / CLS; Deneb (Vega-Lite); Streamlit; Superset

Open-Source & AI: Iceberg; Trino; MinIO; ClickHouse; Airflow; Dagster; Airbyte; OpenMetadata; BigQuery; OpenAI / Claude API; Azure OpenAI; MCP server design; prompt engineering; RAG patterns; AI-assisted engineering

EDUCATION & CERTIFICATIONS

UEH — University of Economics Ho Chi Minh City

Ho Chi Minh City, Vietnam

Certifications: Microsoft DP-700 (Fabric Data Engineer Associate); HackerRank SQL (Advanced); Google Data Analytics

Languages: Vietnamese (native); English (TOEIC 780, daily cross-border collaboration)